

INNOMATRICS TECHNOLOGIES

CRM / ERP – MASTER REQUIREMENTS & DEVELOPMENT SPECIFICATION

Document Type: Master Business + Functional + Technical Development Requirement

Status: Developer Reference / System Blueprint

Company: Innomatrix Technologies

System: Internal CRM + ERP + Business Management Platform

1. PURPOSE OF THIS DOCUMENT

This document is the master requirement specification for developing the internal CRM/ERP platform for Innomatrix Technologies.

The system must not be developed as a simple CRM.

It must become a centralized business management platform covering:

- Leads
- Sales
- Clients
- Quotations
- Agreements
- Projects
- Tasks
- Employees
- HR
- Attendance
- Leave
- Payroll
- Websites
- Mobile applications
- Domains

- Hosting
- Servers
- SSL
- AMC
- Maintenance
- Support
- Finance
- Invoices
- Payments
- Expenses
- Vendors
- Procurement
- Assets
- Software licenses
- API credentials
- Source-code repositories
- App Store / Play Store
- Contracts
- Documents
- Notifications
- Automation
- Client portal
- Employee portal
- Management dashboard
- Reports

- Audit
- Security
- Future AI

The complete business lifecycle should be connected:

Lead → Qualification → Follow-up → Quotation → Negotiation → Approval → Client → Agreement → Project → Development → Testing → Deployment → Support → AMC → Renewal → Payment → Profitability → Retention

The existing blueprint explicitly identifies this complete lifecycle and the broader business areas to be managed.

2. CORE DEVELOPMENT PRINCIPLE

Developers must follow this principle:

Do not build isolated modules. Build one connected business system.

Example:

A lead converted to a client must automatically connect to:

- Client master
- Contact persons
- Quotations
- Agreements
- Projects
- Invoices
- Payments
- Domains
- Hosting
- Applications
- Support
- AMC

- Renewals
- Documents
- Communication history

The same client should not need to be created again in another module.

3. SYSTEM ARCHITECTURE

The application should be organized into the following layers.

Layer 1 – MANAGEMENT

- CEO Dashboard
- Management Dashboard
- Reports
- Analytics
- Business KPIs
- Profitability
- Collections
- Team performance
- Alerts

Layer 2 – CRM

- Leads
- Follow-ups
- Sales pipeline
- Clients
- Contacts
- Quotations
- Proposals

- Agreements
- Sales activities

Layer 3 – OPERATIONS

- Projects
- Tasks
- Milestones
- Development
- Websites
- Mobile applications
- Support
- Change requests
- Deployments
- Warranty

Layer 4 – RECURRING BUSINESS

- Domains
- Hosting
- Servers
- SSL
- AMC
- Maintenance
- SEO
- Digital marketing
- Subscriptions
- Renewals

Layer 5 – FINANCE

- Quotations
- Invoices
- Payments
- Receivables
- Payables
- Expenses
- Purchases
- Vendors
- GST
- Profitability

Layer 6 – PEOPLE & COMPANY

- Employees
- HR
- Attendance
- Leave
- Payroll
- Assets
- Documents
- Vendors
- Procurement

Layer 7 – PLATFORM & SECURITY

- Users
- Roles
- Permissions
- Audit logs

- Security
- API
- Integrations
- Backups
- Settings
- Notifications
- System configuration

Layer 8 – AUTOMATION

All major modules should connect to an automation engine.

Automation channels:

- Email
- WhatsApp
- In-app notifications
- Push notifications
- Tasks
- Follow-ups
- Escalations
- Approvals

Layer 9 – PORTALS

- Client Portal
- Employee Portal
- Future Mobile Application

4. USER ROLES

The system must support role-based access control.

Initial roles should include:

1. Super Admin
2. CEO / Owner
3. Management
4. Admin
5. Sales Manager
6. Sales Executive
7. Project Manager
8. Developer
9. Designer
10. QA
11. Support Executive
12. Finance Manager
13. Accountant
14. HR Manager
15. HR Executive
16. Purchase Manager
17. Employee
18. Client

Admins must be able to create additional roles.

5. ROLE & PERMISSION SYSTEM

Permissions must be configurable.

Permission levels:

- View

- Create
- Edit
- Delete
- Approve
- Export
- Download
- Assign
- Change status
- View financial data
- View confidential data
- Manage users
- Manage settings

Permissions should support:

Module → Submodule → Action

Example:

Sales Executive:

- View leads: Yes
- Create lead: Yes
- Edit own leads: Yes
- View all leads: Depends on permission
- Delete leads: No
- Approve quotation: No
- View company profit: No

CEO:

- Full access

6. LOGIN & ACCOUNT SECURITY

Required:

- Email/username login
- Password
- Forgot password
- Password reset
- Session management
- Login history
- Failed login tracking
- Account lockout
- Optional 2FA
- Device/session tracking
- Logout from all devices
- Password policy
- User activation/deactivation

7. COMPANY-WIDE ACTION CENTER

Create a centralized **Action Center**.

It should display all pending actions.

Examples:

- Follow-up due
- Payment collection
- Domain renewal
- Hosting renewal

- SSL renewal
- AMC renewal
- Project deadline
- Task overdue
- Support SLA risk
- Client approval
- Employee approval
- Purchase approval
- Expense approval
- Quotation approval
- Contract expiry
- Backup failure
- Security alert
- Change request
- Deployment approval

Each action should contain:

- Action ID
- Type
- Owner
- Department
- Client
- Project
- Priority
- Due date
- Status

- Escalation level
- Created date
- Completed date
- Notes
- Related record

Management should be able to see:

“What is pending in the company today?”

from one screen.

8. WORKFLOW ENGINE

Do not hard-code every workflow.

Build a configurable workflow engine.

Structure:

TRIGGER → CONDITION → ACTION → WAIT → REMINDER → ESCALATION

Supported triggers:

- Record created
- Status changed
- Due date approaching
- Payment overdue
- Renewal approaching
- Task overdue
- SLA approaching
- Approval requested
- Approval rejected
- Project delayed

Actions:

- Create task
- Assign employee
- Send email
- Send WhatsApp
- Send notification
- Change status
- Create follow-up
- Create renewal
- Request approval
- Escalate
- Wait
- Create ticket

Admins should be able to configure workflows without code wherever practical.

9. CRM – LEAD MANAGEMENT

Lead fields:

- Lead ID
- Company name
- Contact name
- Phone
- WhatsApp
- Email
- Website
- Location

- Industry
- Lead source
- Service interested
- Estimated budget
- Expected value
- Assigned salesperson
- Lead status
- Priority
- Created date
- Last contacted
- Next follow-up
- Notes

Lead sources:

- Website
- Google Ads
- Facebook
- Instagram
- WhatsApp
- Referral
- Upwork
- Freelancer
- Cold call
- Existing client
- Other

Lead statuses:

- New
 - Contacted
 - Qualified
 - Proposal
 - Negotiation
 - Won
 - Lost
 - On hold
-

10. LEAD FOLLOW-UP

Every lead should support:

- Follow-up date
- Follow-up time
- Follow-up type
- Notes
- Outcome
- Next follow-up
- Assigned employee

Follow-up history must be permanent.

The system should alert users when follow-ups are due.

11. SALES PIPELINE

Pipeline stages should be configurable.

Example:

New Lead



Contacted

↓

Qualified

↓

Requirement Received

↓

Quotation

↓

Negotiation

↓

Approval

↓

Won / Lost

Dashboard:

- Leads by stage
- Lead value
- Conversion rate
- Salesperson performance
- Lost leads
- Pending follow-ups

12. CLIENT MASTER

Client fields:

- Client ID
- Company name
- Contact person
- Email
- Phone
- WhatsApp
- Address

- City
- State
- Country
- GST number
- PAN where applicable
- Industry
- Client type
- Assigned account manager
- Salesperson
- Status
- Notes

Client profile must show a complete timeline.

Example:

Client → Leads → Calls → Emails → Quotations → Agreements → Projects → Websites → Apps → Domains → Hosting → Tickets → Invoices → Payments → AMC → Renewals

13. CLIENT CONTACTS

A client can have multiple contacts.

Fields:

- Contact name
- Designation
- Email
- Phone
- WhatsApp
- Department
- Primary contact

- Billing contact
 - Technical contact
 - Decision maker
 - Status
-

14. QUOTATION / PROPOSAL MODULE

Quotation fields:

- Quotation number
- Client
- Date
- Valid until
- Salesperson
- Project
- Description
- Items
- Quantity
- Rate
- Discount
- Tax
- Total
- Terms
- Payment schedule
- Notes

Quotation statuses:

- Draft

- Pending approval
- Sent
- Viewed
- Negotiation
- Accepted
- Rejected
- Expired
- Cancelled

Generate:

- PDF
 - Email
 - Client portal view
-

15. QUOTATION APPROVAL

For quotations above configured amounts, approval may be required.

Example:

Salesperson creates quotation

↓

Manager approval

↓

CEO approval if required

↓

Quotation sent to client

Maintain:

- Approver
- Date
- Time
- Decision

- Comment
-

16. AGREEMENT / CONTRACT MANAGEMENT

Manage:

- Service agreements
- Development agreements
- AMC agreements
- NDA
- Employment agreements
- Vendor agreements
- Other contracts

Fields:

- Contract number
- Client/vendor
- Contract type
- Start date
- End date
- Value
- Renewal
- Notice period
- Status
- Document
- Responsible employee

Automatic reminders before expiry.

17. DIGITAL APPROVAL / E-SIGNATURE

Where integrated, support:

- Approval request
- Document signing
- Signature status
- Signed date
- Signer
- Signed document
- Audit trail

Do not treat a typed name as a secure signature unless the selected signing mechanism legally/technically supports that requirement.

18. CLIENT ONBOARDING

After sales conversion:

Create onboarding checklist.

Example:

- Client details
- Contact verification
- Agreement
- Advance payment
- Requirements
- Documents
- Credentials
- Project creation
- Team assignment
- Kickoff meeting

Onboarding status:

- Pending
 - In progress
 - Completed
-

19. CLIENT OFFBOARDING

Checklist:

- Pending payments
- Source code
- Hosting
- Domain
- Credentials
- Documents
- Backup
- Final delivery
- Contract closure
- AMC closure
- Access removal

Record who completed each item.

20. PROJECT MANAGEMENT

Project fields:

- Project ID
- Project name
- Client

- Service
- Start date
- Expected end date
- Actual end date
- Project manager
- Team
- Budget
- Quotation
- Contract
- Status
- Priority
- Technology
- Repository
- Hosting
- Deployment details

Project statuses:

- Planning
 - Active
 - On hold
 - Delayed
 - Testing
 - Ready for deployment
 - Completed
 - Cancelled
-

21. PROJECT TASK MANAGEMENT

Task fields:

- Task ID
- Project
- Task name
- Description
- Assigned employee
- Priority
- Start date
- Due date
- Status
- Estimated hours
- Actual hours
- Dependency
- Attachment
- Comments

Statuses:

- To Do
- In Progress
- Blocked
- Review
- QA
- Completed

22. MILESTONES

Each project can contain milestones.

Example:

1. Requirement
2. UI/UX
3. Development
4. Testing
5. Client UAT
6. Deployment
7. Handover

Milestone tracking must show:

- Planned date
 - Actual date
 - Status
 - Completion %
 - Delays
-

23. BUG MANAGEMENT

Bug fields:

- Bug ID
- Project
- Module
- Description
- Screenshot
- Priority
- Severity

- Assigned developer
- QA
- Status
- Created date
- Resolved date

Statuses:

- Open
- Assigned
- In Progress
- Fixed
- QA
- Reopened
- Closed

24. CHANGE REQUEST MANAGEMENT

A change request must be separate from a bug.

Fields:

- CR number
- Project
- Client
- Requirement
- Reason
- Impact
- Estimated hours
- Additional cost

- Approval
- Status

Workflow:

Request → Estimate → Quotation → Client Approval → Development → Testing → Delivery

No chargeable change should automatically become development work before approval unless explicitly configured.

25. WARRANTY MANAGEMENT

Track:

- Delivery date
- Warranty start
- Warranty end
- Covered items
- Excluded items

Support tickets should automatically identify whether they are under warranty.

26. TIME TRACKING

Employees should be able to record:

- Project
- Task
- Start time
- End time
- Duration
- Description

Support:

- Manual entry

- Timer
- Timesheet approval

Use time data for:

- Project profitability
 - Employee productivity
 - Cost calculation
 - Billing where applicable
-

27. RESOURCE PLANNING

Management should see:

- Employee workload
 - Assigned projects
 - Hours
 - Availability
 - Overloaded employees
 - Underutilized employees
 - Upcoming assignments
-

28. WEBSITE PORTFOLIO

Because Innomatrix manages many websites, every website must have a portfolio record.

Fields:

- Website ID
- Client
- Domain
- Project

- Website type
- Technology
- Framework
- Repository
- Hosting
- Server
- SSL
- Database
- Deployment date
- Renewal date
- AMC
- Status
- Login references
- Backup status

Website types:

- Corporate
 - Business
 - E-commerce
 - Custom web application
 - WordPress
 - PHP
 - React
 - Laravel
 - Other
-

29. MOBILE APPLICATION PORTFOLIO

Fields:

- App ID
- Client
- App name
- Android package
- iOS bundle ID
- Technology
- Flutter/native/etc.
- Firebase project
- Repository
- API
- Server
- Database
- Play Store URL
- App Store URL
- Version
- Current production version
- Last release
- Signing/certificate reference
- Status

30. PLAY STORE / APP STORE MANAGEMENT

Track:

- Developer account

- App
- Package ID
- Bundle ID
- Version
- Build number
- Release date
- Review status
- Production status
- Test status
- Rejection history
- Store credentials reference
- Certificates
- Provisioning profiles
- Screenshots
- Store description
- Privacy policy
- Support URL

Never store highly sensitive secrets as plain text.

31. DOMAIN MANAGEMENT

Fields:

- Domain
- Client
- Registrar
- Registrar account reference

- Registration date
- Expiry date
- Auto-renew
- Nameservers
- DNS provider
- Status
- SSL
- Hosting
- Responsible employee

Renewal alerts:

- 90 days
- 60 days
- 30 days
- 15 days
- 7 days
- Expired

The existing requirements specifically call for automated renewal handling and reminder/escalation workflows.

32. HOSTING MANAGEMENT

Track:

- Hosting provider
- Account
- Server
- Plan
- Client

- Website
 - Start date
 - Expiry date
 - Cost
 - Selling price
 - Renewal
 - IP
 - Control panel
 - Backup
 - Status
-

33. SERVER MANAGEMENT

Track:

- Server name
- Provider
- IP
- Region
- OS
- CPU
- RAM
- Storage
- Environment
- Application
- Client
- Monitoring

- Backup
- Renewal
- Status

Environments:

- Development
 - Staging
 - Production
-

34. SSL MANAGEMENT

Track:

- Domain
 - Certificate type
 - Provider
 - Issue date
 - Expiry date
 - Auto-renew
 - Installation status
 - Responsible employee
-

35. BACKUP MANAGEMENT

Track backups for:

- Websites
- Databases
- Servers
- Source code

- Documents

Fields:

- Backup type
- Source
- Destination
- Frequency
- Last successful backup
- Last failed backup
- Retention
- Status

Automatic alert on backup failure.

36. RECURRING SERVICES

Support:

- Domain
- Hosting
- Server
- SSL
- AMC
- Maintenance
- SEO
- Digital marketing
- Software subscription
- Other recurring services

Fields:

- Client
 - Service
 - Start date
 - Billing cycle
 - Cost
 - Selling price
 - Margin
 - Next billing date
 - Renewal date
 - Auto-renew
 - Payment status
 - Cancellation date
 - Contract
-

37. AMC MANAGEMENT

Track:

- Client
- Project/service
- AMC start
- AMC end
- AMC value
- Coverage
- SLA
- Renewal status
- Payment status

Automatically create renewal opportunity before expiry.

38. SUPPORT / HELPDESK

Ticket fields:

- Ticket ID
- Client
- Contact
- Project
- Service
- Category
- Priority
- Subject
- Description
- Attachment
- Assigned employee
- SLA
- Status
- Created date
- Due date
- Resolution
- Closed date

Statuses:

- Open
- Assigned
- In Progress

- Waiting for Client
 - Resolved
 - Closed
 - Reopened
-

39. SLA MANAGEMENT

SLA configuration:

- Priority
- Response time
- Resolution time
- Working hours
- Escalation level

Alerts:

- SLA approaching
- SLA breached

Management dashboard must show SLA risks.

40. CLIENT COMMUNICATION

Store communication history:

- Email
- WhatsApp
- Calls
- Meetings
- Notes
- Support messages

Each communication should link to:

- Client
 - Contact
 - Lead/project/ticket
 - Employee
 - Date/time
-

41. CLIENT SATISFACTION

After project/support completion:

- Rating
- Feedback
- Comments
- Satisfaction score

Track recurring negative feedback for management review.

42. REFERRAL MANAGEMENT

Track:

- Referrer
 - Referred client
 - Referral date
 - Value
 - Commission if applicable
 - Status
 - Payment
-

43. FINANCE MODULE

Finance should include:

- Quotations
 - Invoices
 - Payments
 - Receivables
 - Payables
 - Expenses
 - Purchases
 - Vendors
 - GST
 - Profitability
-

44. INVOICE MANAGEMENT

Fields:

- Invoice number
- Client
- Project
- Quotation
- Invoice date
- Due date
- Items
- Tax
- Discount
- Total

- Amount paid
- Balance
- Status

Statuses:

- Draft
 - Sent
 - Partially paid
 - Paid
 - Overdue
 - Cancelled
-

45. PAYMENT MANAGEMENT

Payment fields:

- Payment ID
- Client
- Invoice
- Amount
- Date
- Payment method
- Transaction reference
- Bank
- Gateway
- Notes

Payment methods:

- Bank transfer

- UPI
 - Cash
 - Cheque
 - Payment gateway
 - Other
-

46. RECEIVABLES

Dashboard:

- Total receivable
- Due today
- Due this week
- Overdue
- Aging

Aging:

- 0–30 days
- 31–60
- 61–90
- 90+

Automatic collection reminders.

47. PAYABLES

Track:

- Vendor
- Invoice
- Amount

- Due date
 - Payment status
 - Payment date
-

48. EXPENSE MANAGEMENT

Expense fields:

- Expense ID
- Employee
- Category
- Amount
- Date
- Vendor
- Project
- Client
- Receipt
- Description
- Approval status

Workflow:

Employee → Manager → Finance → Paid

49. PROCUREMENT

Purchase request:

- Requester
- Department
- Item/service

- Quantity
- Estimated amount
- Vendor
- Required date
- Reason
- Approval

Workflow:

Request → Approval → Purchase Order → Vendor → Invoice → Payment

50. VENDOR MANAGEMENT

Track:

- Vendor
 - Contact
 - Service
 - GST
 - Bank details
 - Contracts
 - Payments
 - Documents
 - Renewal
 - Performance
-

51. PROJECT PROFITABILITY

For each project calculate:

Revenue – Direct Cost – Employee Cost – Vendor Cost – Infrastructure Cost – Other Cost = Project Profit

Show:

- Estimated profit
 - Actual profit
 - Margin %
 - Budget variance
-

52. CLIENT PROFITABILITY

Calculate client-level:

- Revenue
 - Cost
 - Support cost
 - Employee hours
 - Infrastructure cost
 - AMC revenue
 - AMC cost
 - Total profit
 - Margin
-

53. EMPLOYEE MANAGEMENT / HR

Employee fields:

- Employee ID
- Name
- Email
- Phone
- Department

- Designation
 - Joining date
 - Manager
 - Employment status
 - Salary information
 - Documents
 - Skills
-

54. ATTENDANCE

Support:

- Login
- Logout
- Attendance
- Late arrival
- Early exit
- Work hours
- Overtime
- WFH
- Holiday

Reports:

- Daily
 - Monthly
 - Employee
 - Department
-

55. LEAVE MANAGEMENT

Leave types configurable.

Workflow:

Employee → Manager/HR → Approval → Balance Update

Track:

- Leave balance
 - Leave taken
 - Pending
 - Rejected
 - Holidays
-

56. PAYROLL

Payroll should support:

- Employee
- Basic salary
- Allowances
- Deductions
- Incentives
- Overtime
- Leave deductions
- Net salary
- Payroll month
- Payment status

Payroll access must be highly restricted.

57. EMPLOYEE KPI

KPIs may include:

- Tasks completed
- Tasks overdue
- Project contribution
- Billable hours
- Support tickets
- Response time
- Sales
- Collections
- Attendance
- Client feedback

KPI formulas should be configurable.

58. EMPLOYEE ASSETS

Track:

- Laptop
- Desktop
- Mobile
- SIM
- Monitor
- Accessories
- Software
- Other assets

Fields:

- Asset ID
 - Serial number
 - Employee
 - Issue date
 - Return date
 - Condition
 - Status
-

59. SOFTWARE LICENSE MANAGEMENT

Track:

- Software
 - License type
 - Account
 - User
 - Purchase date
 - Expiry
 - Cost
 - Renewal
 - License key reference
-

60. API CREDENTIAL / SECRET MANAGEMENT

The system should store references to:

- API keys
- Firebase
- AWS

- Hosting
- Database
- Payment gateway
- Email
- WhatsApp
- Third-party services

Sensitive credentials must:

- Be encrypted
- Never appear in normal logs
- Have restricted permissions
- Support rotation
- Maintain access history

Do not store plaintext passwords unnecessarily.

61. SOURCE CODE / REPOSITORY MANAGEMENT

Track:

- Repository
- Provider
- Organization
- Project
- Repository URL
- Main branch
- Development branch
- Staging branch
- Production branch

- Owner
- Access level

Possible integrations:

- GitHub
- GitLab
- Bitbucket

62. DOCUMENT MANAGEMENT

Documents may belong to:

- Client
- Employee
- Project
- Contract
- Vendor
- Invoice
- Asset
- Domain
- Website
- Application

Features:

- Upload
- Download
- Version
- Category
- Access control

- Expiry
 - Search
 - Audit history
-

63. GLOBAL SEARCH

Global search should search:

- Client
- Lead
- Employee
- Project
- Task
- Ticket
- Website
- App
- Domain
- Invoice
- Payment
- Contract
- Document

Search should be fast and scalable.

64. ACTIVITY TIMELINE

Every important record should have a timeline.

Example:

Client ABC

- Created
 - Lead converted
 - Quotation sent
 - Quotation approved
 - Payment received
 - Project created
 - Task assigned
 - Deployment completed
 - Ticket opened
 - AMC renewed
-

65. NOTIFICATION ENGINE

Channels:

- In-app
- Email
- WhatsApp
- Push
- SMS where required

Notification types:

- Follow-up
- Payment
- Renewal
- Task
- Project
- Ticket

- Approval
- Leave
- Expense
- Security
- Backup

Users should control notification preferences where appropriate.

66. AUTOMATIC REMINDERS

Recurring reminder examples:

Domain

90 days
60 days
30 days
15 days
7 days
Expired

AMC

60 days
30 days
15 days
7 days
Expired

Invoice

Before due date
Due date
1 day overdue
7 days overdue
15 days overdue
30 days overdue

Reminder intervals must be configurable.

The existing requirements specifically call for reminders before/on/after due dates and escalation.

67. BUSINESS CALENDAR

Calendar should show:

- Meetings
- Follow-ups
- Tasks
- Deadlines
- Renewals
- Payments
- Leave
- Client approvals
- Contract expiry

68. MEETING MANAGEMENT

Meeting fields:

- Meeting title
 - Client
 - Participants
 - Date/time
 - Meeting type
 - Agenda
 - Notes
 - Action items
 - Next follow-up
-

69. CLIENT PORTAL

Clients should have secure access.

Client can view:

- Profile
- Projects
- Project progress
- Quotations
- Invoices
- Payments
- Support tickets
- Documents
- Agreements
- Renewals
- Approvals
- Change requests

Client should not see internal information unless explicitly permitted.

70. CLIENT APPROVALS

Client should be able to approve:

- Quotation
- Requirement
- Design
- UAT
- Change request
- Deployment

- Final delivery

Record:

- User
 - Date
 - Time
 - IP/device information where appropriate
 - Comment
 - Decision
-

71. EMPLOYEE PORTAL

Employee can access:

- Dashboard
 - Tasks
 - Projects
 - Attendance
 - Leave
 - Expenses
 - Assets
 - Documents
 - Meetings
 - Notifications
 - Approvals
-

72. MANAGEMENT DASHBOARD

Management dashboard should include:

Sales

- New leads
- Qualified leads
- Open opportunities
- Sales value
- Won
- Lost

Finance

- Revenue
- Collections
- Receivables
- Payables
- Expenses
- Profit

Operations

- Active projects
- Delayed projects
- Tasks
- Bugs
- Change requests
- Support tickets

Recurring

- Domains
- Hosting
- SSL

- AMC
- Renewals

Team

- Employees
 - Attendance
 - Leave
 - Workload
 - Overdue tasks
-

73. CEO MORNING DASHBOARD

Create a dedicated CEO view.

NEEDS ATTENTION

- Overdue payments
- Domain renewals
- Delayed projects
- Unanswered leads
- SLA risks
- Pending approvals
- Security alerts
- Backup failures

COMING UP

- Renewals this month
- AMC expiries
- Contracts expiring
- Receivables due

- Project deadlines

BUSINESS

- New leads
- New sales
- Collections
- Active projects
- Recurring revenue
- Expenses
- Profit

TEAM

- Overloaded employees
- Overdue tasks
- Pending approvals
- Leave
- Attendance

This “CEO Morning View” is specifically identified in the source requirements as a key management screen.

74. DASHBOARD WIDGET SYSTEM

Widgets should be configurable.

Examples:

- Leads
- Sales
- Revenue
- Collections
- Projects

- Tasks
- Tickets
- Renewals
- Expenses
- Employees
- Attendance
- Profitability

Users can configure their dashboard based on permissions.

75. REPORTING SYSTEM

Reports should support:

- Date filter
- Client filter
- Employee filter
- Department filter
- Project filter
- Status
- Service
- Export

Export:

- Excel
- CSV
- PDF

Reports should respect user permissions.

76. REQUIRED REPORTS

Minimum reports:

Sales

- Lead report
- Conversion report
- Salesperson report
- Lost lead report

Finance

- Revenue
- Invoice
- Payment
- Receivable
- Payable
- Expense
- Profit

Operations

- Project
- Task
- Delay
- Employee workload
- Support
- SLA

Recurring

- Domain renewal
- Hosting renewal

- SSL expiry
- AMC expiry
- Subscription

HR

- Attendance
 - Leave
 - Payroll
 - Employee KPI
-

77. DATA MIGRATION

Existing business data must be imported.

Potential records:

- Clients
- Websites
- Mobile applications
- Domains
- Hosting
- Projects
- Employees
- Vendors
- Contracts
- Documents
- AMC
- Payments

Migration process:

Import → Validate → Duplicate Check → Preview → Approve → Import → Report

Migration logs must be maintained.

Special requirement:

Existing 200+ websites/applications should be capable of being imported into the new system.

78. DUPLICATE DETECTION

Before creating records, check duplicates.

Client duplicate checks:

- Email
- Phone
- Company
- GST

Domain:

- Domain name

Employee:

- Email
- Employee ID

Invoice:

- Invoice number
-

79. CUSTOM FIELDS

Admins should be able to create custom fields without database code changes where practical.

Examples:

Client:

- Industry

Project:

- Technology version

Domain:

- Registrar account

Lead:

- Budget range

The source requirements explicitly identify configurable custom fields as essential.

80. CUSTOM STATUSES

Admins should be able to configure statuses for:

- Leads
- Projects
- Tickets
- Payments
- Renewals
- Tasks
- Approvals

Do not hard-code every status.

81. AUDIT LOG

Record important actions:

- Login
- Logout
- Create
- Update

- Delete
- Status change
- Approval
- Rejection
- Payment
- Export
- Document download
- Permission change
- Credential access

Audit record:

- User
 - Action
 - Record
 - Old value
 - New value
 - Date/time
 - IP/device where appropriate
-

82. SOFT DELETE

Important business records should not be permanently deleted by normal users.

Use:

- Active
- Archived
- Deleted/soft deleted

Admin recovery should be available where appropriate.

83. DATA SECURITY

Required:

- HTTPS
- Secure password hashing
- Encryption for sensitive data
- Secure API authentication
- RBAC
- Session security
- Rate limiting
- Input validation
- SQL injection protection
- XSS protection
- CSRF protection where applicable
- Secure file upload
- File type validation
- Access logging

84. DATA RETENTION & PRIVACY

Define:

- Access rights
- Retention period
- Document access
- Export rights
- Audit history

- Backup retention
- Deletion procedures

These policies should be configurable according to applicable legal and contractual requirements.

85. API ARCHITECTURE

The CRM/ERP must expose secure APIs.

API areas:

- Authentication
- Users
- Clients
- Leads
- Sales
- Quotations
- Projects
- Tasks
- Tickets
- Finance
- Domains
- Hosting
- Websites
- Apps
- Employees
- Attendance
- Leave
- Notifications

- Documents
- Reports

The source requirements explicitly call for APIs for website, mobile app, accounting, payment, email, WhatsApp, monitoring, calendar, storage and analytics integrations.

86. THIRD-PARTY INTEGRATIONS

Architecture should allow integration with:

- Company website
- Payment gateway
- Email provider
- WhatsApp provider
- SMS provider
- Calendar
- Cloud storage
- GitHub/GitLab/Bitbucket
- Accounting software
- Cloud/server monitoring
- Analytics
- App Store
- Play Store

Each integration should have:

- API configuration
- Credentials
- Connection status
- Logs
- Error handling

- Retry mechanism
-

87. EMAIL SYSTEM

Support:

- SMTP
- Transactional emails
- Templates
- Notifications
- Quotations
- Invoices
- Payment reminders
- Renewal reminders
- Ticket notifications

Email templates should be configurable.

88. WHATSAPP SYSTEM

If a WhatsApp Business/API provider is connected, support:

- Lead follow-up
- Appointment reminders
- Quotation notification
- Invoice reminder
- Payment confirmation
- Ticket updates
- Renewal reminders

All automated messages must be logged.

89. FILE STORAGE

Store documents separately from database records.

Database should store:

- File ID
- File name
- File type
- Storage location
- Owner
- Related record
- Upload date
- Version
- Access permissions

Support large files safely.

90. SEARCH ARCHITECTURE

Global search should be optimized for large data.

Search:

- Client
- Lead
- Project
- Task
- Ticket
- Domain
- Website

- App
- Invoice
- Payment
- Employee
- Document

Use indexing/search technology appropriate for expected scale.

91. PERFORMANCE REQUIREMENTS

System should support:

- Pagination
- Lazy loading
- Efficient queries
- Database indexing
- Caching
- Background jobs
- Queue processing
- Large file handling
- Search optimization
- API optimization

Do not load thousands of records unnecessarily.

The existing blueprint specifically calls for fast dashboard loading, pagination, efficient queries, background jobs, caching and scalable APIs.

92. BACKGROUND JOB SYSTEM

Use background jobs for:

- Emails

- WhatsApp
- Notifications
- Reminders
- Renewal creation
- Report generation
- Backup checks
- Scheduled workflows
- Large imports

Users should not wait for long background operations.

93. QUEUE & RETRY

External operations should support:

- Queue
- Retry
- Failure status
- Error log
- Retry count
- Manual retry

Example:

Email failed

↓

Retry

↓

Still failed

↓

Log error

↓

Notify admin

94. MULTI-COMPANY / MULTI-BRANCH READINESS

Even if only one company exists initially, database architecture must allow:

- Company
- Branch
- Division

Future structure:

Innomatrics Technologies

→ Bengaluru

→ Future Branch

→ Future Division

→ Future Company

The source blueprint explicitly requests this readiness without requiring activation in Phase 1.

95. DATABASE DESIGN

Developers must create a normalized database structure.

Major entity groups:

Identity

- users
- roles
- permissions
- role_permissions
- user_roles
- sessions

CRM

- leads
- lead_sources
- lead_activities

- followups
- clients
- client_contacts
- sales_opportunities

Sales

- quotations
- quotation_items
- quotation_approvals
- contracts
- contract_documents

Projects

- projects
- project_members
- milestones
- tasks
- task_comments
- task_dependencies
- bugs
- change_requests
- deployments

Digital Assets

- websites
- applications
- domains
- hosting

- servers
- ssl_certificates
- repositories
- app_store_records

Support

- tickets
- ticket_messages
- ticket_attachments
- sla_rules

Finance

- invoices
- invoice_items
- payments
- expenses
- expense_items
- purchase_requests
- purchase_orders
- vendors
- receivables
- payables

HR

- employees
- departments
- designations
- attendance

- leave_types
- leave_requests
- payroll
- employee_kpis

Assets

- assets
- asset_assignments
- software_licenses

Documents

- documents
- document_versions
- document_permissions

Automation

- workflows
- workflow_triggers
- workflow_conditions
- workflow_actions
- scheduled_jobs
- notifications

Audit

- audit_logs
- login_logs
- security_events

Developers must prepare an ERD before final database implementation.

96. COMMON DATABASE FIELDS

Where applicable, standard fields should include:

- id
- uuid
- company_id
- branch_id
- created_by
- updated_by
- created_at
- updated_at
- deleted_at
- status

Use consistent naming conventions throughout the system.

97. API RESPONSE STANDARD

All APIs should use a consistent structure.

Example:

```
{  
  "success": true,  
  "message": "Client created successfully",  
  "data": {},  
  "errors": []  
}
```

Errors should provide safe, useful messages.

Never expose:

- Database errors
- Passwords
- API keys
- Internal secrets
- Stack traces

in production responses.

98. APPROVAL ENGINE

Create a reusable approval system.

Used for:

- Quotations
- Expenses
- Purchases
- Leave
- Discounts
- Change requests
- Deployments
- Contracts
- Payments

Approval flow:

Created → Pending Approval → Approved / Rejected

Support multiple levels.

Example:

₹0–₹25,000 → Manager

₹25,001–₹1,00,000 → Manager + Finance

Above ₹1,00,000 → CEO

Thresholds must be configurable.

99. USER CUSTOMIZATION

Where practical, users can configure:

- Dashboard layout
- Table columns
- Saved filters
- Saved reports
- Default views
- Notification preferences

This requirement is already included in the blueprint.

100. AI ASSISTANT – FUTURE PHASE

AI should be implemented only after the CRM/ERP core is stable.

Possible questions:

- Which renewals are due this week?
- Which clients have overdue payments?
- What happened with a specific client?
- Which projects are delayed?
- Prepare tomorrow's follow-up list.
- Summarize client history.
- Which projects are over budget?
- Which AMC contracts expire this month?

AI rules:

1. Respect RBAC.

2. Use authorized CRM data only.
3. Do not invent records.
4. Show source records where practical.
5. Log AI queries where appropriate.
6. Protect confidential information.

The source requirements explicitly specify permission-aware AI and prohibit invented CRM information.

101. FUTURE MOBILE APPLICATION

Future mobile application may provide:

- Dashboard
- Tasks
- Notifications
- Leads
- Clients
- Tickets
- Attendance
- Leave
- Approvals
- Meetings
- Push notifications

Mobile app must use the same secure backend APIs.

102. SYSTEM SETTINGS

Admin settings should include:

- Company

- Branch
- Currency
- Tax
- Numbering
- Email
- WhatsApp
- Notifications
- Workflow
- Approval
- Roles
- Permissions
- Custom fields
- Statuses
- Categories
- Departments
- Designations
- Leave types
- Expense categories
- Ticket categories
- Project categories

103. NUMBERING SYSTEM

Configurable numbering:

- Lead
- Client

- Quotation
- Invoice
- Project
- Task
- Ticket
- Contract
- Expense
- Purchase
- Payment
- Change request

Example:

INN-LEAD-000001

INN-QUO-000001

INN-INV-000001

INN-PROJ-000001

104. AUDITABLE BUSINESS HISTORY

Important records must maintain history.

Do not overwrite important business history without logging the change.

Example:

Quotation ₹80,000

changed to ₹75,000

System must record:

- Previous amount
- New amount
- Changed by
- Date

- Reason if configured
-

105. CLIENT RETENTION / RENEWAL

System should automatically identify:

- AMC expiry
- Hosting expiry
- Domain expiry
- Contract expiry
- Maintenance renewal
- Subscription renewal

Create renewal opportunities automatically where configured.

106. AUTOMATED REMINDER ENGINE

Support:

- Before due date
- On due date
- After due date
- Repeated reminders
- Escalation

Example:

Domain expiry:

90 days → reminder

60 days → reminder

30 days → reminder

15 days → reminder

7 days → urgent

Expired → escalation

107. ESCALATION ENGINE

Escalate when:

- Follow-up overdue
- Payment overdue
- Task overdue
- SLA breached
- Renewal ignored
- Approval pending
- Project delayed
- Backup failed

Escalation chain should be configurable.

108. DASHBOARD WIDGET SYSTEM

Widgets should be permission-based and configurable.

Possible widgets:

- Leads
- Sales
- Revenue
- Collections
- Projects
- Tasks
- Tickets
- Renewals
- Expenses

- Employees
 - Attendance
 - Profitability
-

109. USER CUSTOMIZATION

Allow:

- Dashboard layout
 - Notification preferences
 - Table columns
 - Saved filters
 - Saved reports
 - Default views
-

110. SYSTEM PERFORMANCE

System should be designed for:

- Fast dashboard loading
- Pagination
- Efficient database queries
- Background jobs
- Queue processing
- Caching
- Large file handling
- Search optimization
- Scalable APIs

Avoid loading unnecessary data.

111. ERROR HANDLING

Application must have:

- Centralized error handling
- User-friendly messages
- Developer logs
- API error logs
- Integration error logs
- Background-job failure logs

Production must not expose technical stack traces.

112. SYSTEM LOGGING

Logs should cover:

- Application errors
- API errors
- Login
- Security events
- Background jobs
- Integration failures
- Email failures
- Notification failures
- Backup failures

Log retention should be configurable.

113. ENVIRONMENT MANAGEMENT

Minimum environments:

Development

Developer testing.

Staging

QA/UAT.

Production

Live system.

Never directly test major changes in production.

Use environment variables for:

- Database credentials
- API keys
- Email
- Storage
- Payment gateway
- WhatsApp
- Other secrets

114. BACKUP & DISASTER RECOVERY

Define:

- Database backup frequency
- File backup
- Off-site backup
- Retention
- Restore process
- Recovery testing

- Disaster recovery plan

Backup failure must create an alert.

115. TESTING REQUIREMENTS

Testing must cover:

Unit testing

Individual functions/modules.

API testing

All APIs.

Integration testing

Module-to-module workflows.

Security testing

Authentication, authorization, access control.

Performance testing

Dashboard/search/API performance.

User acceptance testing

Actual business workflows.

116. UAT CHECKLIST

Before production:

- Login
- Roles
- Permissions
- Leads
- Follow-ups

- Clients
- Quotations
- Approvals
- Projects
- Tasks
- Websites
- Apps
- Domains
- Hosting
- Tickets
- Invoices
- Payments
- Expenses
- HR
- Attendance
- Leave
- Reports
- Notifications
- Client portal
- Employee portal
- Audit
- Backup
- Data import

Every module must have test cases and acceptance criteria.

117. FINAL SYSTEM ARCHITECTURE

The final system should follow:

MANAGEMENT

Dashboard • Reports • Analytics • CEO View

|



CRM

Leads • Sales • Clients • Follow-ups

|



OPERATIONS

Projects • Tasks • Websites • Apps • Support

|



RECURRING BUSINESS

Domains • Hosting • SSL • AMC • Renewals

|



FINANCE

Quotes • Invoices • Payments • Expenses

Receivables • Payables • GST • Profit

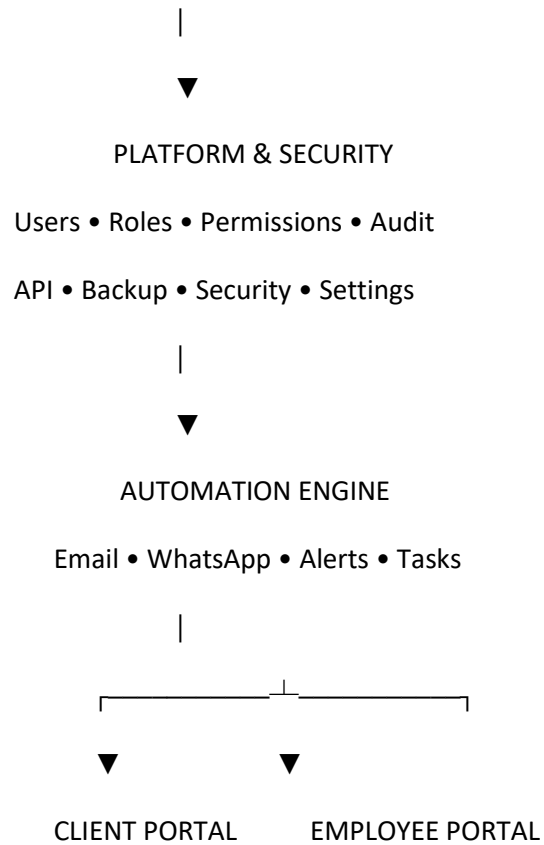
|



PEOPLE & COMPANY

HR • Attendance • Leave • Payroll

Assets • Vendors • Procurement • Documents



This architecture follows the six connected business layers defined in the original requirements, with the platform/security and automation layers made explicit for development.

118. MASTER BUSINESS FLOW

LEAD



QUALIFICATION



FOLLOW-UP



PROPOSAL / QUOTATION



NEGOTIATION



APPROVAL



CLIENT



AGREEMENT



ADVANCE PAYMENT



PROJECT



REQUIREMENTS



DESIGN



DEVELOPMENT



QA / TESTING



CLIENT UAT



DEPLOYMENT



FINAL PAYMENT



HANDOVER



WARRANTY



SUPPORT



AMC / MAINTENANCE



RENEWAL



RETENTION

Every stage must create/maintain the required connected records.

119. CLIENT 360° VIEW

A client page must show:

CLIENT



├— Contacts

├— Leads

├— Sales

├— Quotations

├— Agreements

├— Projects

├— Tasks

├— Websites

- └─ Mobile Apps
- └─ Domains
- └─ Hosting
- └─ SSL
- └─ AMC
- └─ Support Tickets
- └─ Invoices
- └─ Payments
- └─ Expenses
- └─ Documents
- └─ Meetings
- └─ Communications
- └─ Renewals
- └─ Activity Timeline

This should be one of the most important screens in the system.

120. PROJECT 360° VIEW

Project page:

PROJECT

- |
- └─ Client
- └─ Quotation
- └─ Contract
- └─ Team
- └─ Requirements

└─ Milestones

└─ Tasks

└─ Bugs

└─ Change Requests

└─ Time

└─ Expenses

└─ Repository

└─ Website/App

└─ Hosting

└─ Deployment

└─ Documents

└─ Payments

└─ Profitability

└─ Timeline

121. DEVELOPMENT PHASES

PHASE 1 – FOUNDATION

- Architecture
- Database
- Authentication
- RBAC
- Company
- Users
- Audit
- Settings

- Notifications

PHASE 2 – CRM

- Leads
- Follow-ups
- Sales pipeline
- Clients
- Contacts
- Quotations
- Approvals
- Contracts

PHASE 3 – PROJECTS

- Projects
- Tasks
- Milestones
- Bugs
- Change requests
- Time tracking
- Resource management

PHASE 4 – DIGITAL ASSETS

- Websites
- Applications
- Domains
- Hosting
- Servers
- SSL

- Repositories
- Store records

PHASE 5 – SUPPORT

- Tickets
- SLA
- Warranty
- Client communication

PHASE 6 – FINANCE

- Invoices
- Payments
- Receivables
- Payables
- Expenses
- Procurement
- Vendors
- Profitability
- GST

PHASE 7 – HR

- Employees
- Attendance
- Leave
- Payroll
- KPI
- Assets

PHASE 8 – PORTALS

- Client portal
- Employee portal

PHASE 9 – AUTOMATION

- Workflow engine
- Reminders
- Escalation
- Email
- WhatsApp
- Scheduled jobs

PHASE 10 – ADVANCED

- Analytics
- AI
- Mobile application
- Advanced integrations

122. DEVELOPER DELIVERABLES

Development team must deliver:

1. UI/UX design
2. Database ERD
3. Database schema
4. Backend/API
5. Web application
6. Authentication
7. RBAC
8. All modules

9. Client portal
 10. Employee portal
 11. Notification system
 12. Workflow engine
 13. Reports
 14. Audit system
 15. API documentation
 16. Technical documentation
 17. Deployment documentation
 18. Backup documentation
 19. Test cases
 20. UAT documentation
 21. Source code
 22. Environment configuration documentation
 23. Database migration scripts
 24. Deployment scripts where applicable
-

123. DEVELOPMENT RULES

Developers must follow these rules:

Rule 1

Do not duplicate client data between modules.

Rule 2

Do not hard-code configurable business rules unnecessarily.

Rule 3

All important actions must be auditable.

Rule 4

All APIs must have authentication and authorization.

Rule 5

Sensitive credentials must never be stored or displayed insecurely.

Rule 6

Do not expose confidential data to unauthorized users.

Rule 7

Use database transactions for critical financial operations.

Rule 8

Use background jobs for long-running operations.

Rule 9

Every module must support proper validation.

Rule 10

Every major module must have search, filters and pagination.

Rule 11

Every important record must have status and history.

Rule 12

All production changes must be tested before deployment.

Rule 13

Do not directly modify production data without authorization.

Rule 14

Maintain proper Git/version control.

Rule 15

Database migrations must be version controlled.

124. DEVELOPER SCREEN REQUIREMENT

For every screen, developers must document:

**Module → Screen → Fields → Actions → Workflow → Permissions → Database → API → Notifications
→ Reports → Test Cases**

Example:

Client Creation Screen

Fields

- Company
- Contact
- Email
- Phone
- GST
- Address
- Industry
- Source
- Account manager

Actions

- Save
- Save & Create Project
- Cancel

Permissions

- Admin: Full
- Sales: Create/Edit
- Developer: View
- Client: No

API

POST /api/clients

Database

clients

Notifications

Notify account manager after creation.

Audit

Record creation.

125. ACCEPTANCE CRITERIA

A module is not considered complete merely because the screen works.

Each module must have:

- UI
- Backend
- Database
- API
- Validation
- Permissions
- Audit
- Notifications
- Search
- Filters
- Pagination
- Error handling
- Test cases
- UAT approval

126. FINAL DEVELOPMENT CHECKLIST

Before declaring the system complete:

SECURITY

- Authentication
- RBAC
- Password security
- Session security
- API security
- Audit
- Encryption
- Backup
- Restore testing

CRM

- Leads
- Follow-ups
- Sales
- Clients
- Contacts
- Quotations
- Contracts

OPERATIONS

- Projects
- Tasks
- Milestones

- Bugs
- Change requests
- Deployment
- Warranty
- Support
- SLA

DIGITAL ASSETS

- Websites
- Mobile apps
- Domains
- Hosting
- Servers
- SSL
- Repositories
- Play Store
- App Store

FINANCE

- Invoices
- Payments
- Receivables
- Payables
- Expenses
- Procurement
- Vendors
- GST

- Profitability

HR

- Employees
- Attendance
- Leave
- Payroll
- KPI
- Assets

AUTOMATION

- Workflow
- Reminders
- Escalation
- Email
- WhatsApp
- Notifications
- Scheduled jobs

PORTALS

- Client portal
- Employee portal

MANAGEMENT

- CEO dashboard
- Management dashboard
- Reports
- Analytics
- Action center

DATA

- Migration
- Duplicate detection
- Import validation
- Existing 200+ websites/apps
- Documents
- Global search

FUTURE

- AI assistant
- Mobile application
- Advanced integrations
- Multi-company
- Multi-branch

127. FINAL INSTRUCTION TO DEVELOPMENT TEAM

This document is the **master business and development requirement** for the Innomatrix Technologies CRM/ERP.

Developers must not begin full implementation based only on individual screens or verbal instructions.

Before development of each module, create and obtain internal approval for:

1. UI/UX
2. Screen list
3. Field list
4. Database tables
5. ERD relationships
6. API specification
7. Role/permission matrix

8. Workflow
9. Approval rules
10. Notification rules
11. Audit requirements
12. Reports
13. Test cases
14. UAT criteria

The system must be designed as a **scalable, secure, modular business platform**, not a collection of disconnected CRUD screens.

The primary objective is:

One system where management can understand the complete state of the company — sales, clients, projects, employees, digital assets, support, renewals, finance and profitability — without depending on separate spreadsheets or asking multiple employees for updates.

The system should remain extensible for future:

- AI
- Mobile applications
- Additional branches
- Additional companies
- Additional integrations
- Advanced analytics
- Automation